Age at Exposure to Violence

Assessment of the Colombia design

Job market paper, idea 1

October 2026

The idea

Does the effect of violence on adult human capital depend on the age at which the child is exposed?

- Setting: Colombian paramilitary violence, 1997–2006. Entry staggered across municipalities, a ceasefire in December 2002, 38 bloc demobilizations between 2003 and 2006, zero paramilitary victims after 2007.
- Variation: children born in the same municipality were of different ages when the violence ended.
- Outcomes: schooling, employment, earnings at ages 20–36 (GEIH), with municipality of birth.
- Mechanism: household channel (caregiver stress, displacement) versus the child's direct exposure.

What we would have estimated

$$Y_{imc} = \sum_a \beta_a \text{ Years exposed}^a_{mc} + \mu_m + \lambda_c + \text{trend}_{d(m)}(c) + \varepsilon_{imc}$$

- $a =$ in utero, early childhood, preschool, primary, high school.
- μ_m: municipality-of-birth fixed effects. λ_c: year-of-birth fixed effects. Province trends.
- Validity: violence before conception must have no effect.

This is, line by line, equation (3) of León (2012, *JHR*).

The question is already well covered

Paper	Shock	Age windows
Bundervoet, Verwimp and Akresh (2009, *JHR*)	Burundi civil war	months of exposure
Akresh, Bhalotra, Leone and Osili (2012, *AER P&P*)	Biafran war	birth to adolescence
León (2012, *JHR*)	Shining Path, Peru	in utero to high school
Akbulut-Yuksel (2014, *JHR*)	WWII bombing, Germany	school age
Singhal (2019, *JDE*)	US bombing, Vietnam	early life
Duque (2024, *JCR*)	Colombia, 1980s homicides	in utero to adolescence
Akbulut-Yuksel, Tekin and Turan (2025, *JHR*)	WWII bombing, Germany	≤ 5 vs 6–10

Same design in all of them: birth location $\times$ cohort, fixed effects, exposure by age window. A fourth or fifth paper of this kind is not a job market paper.

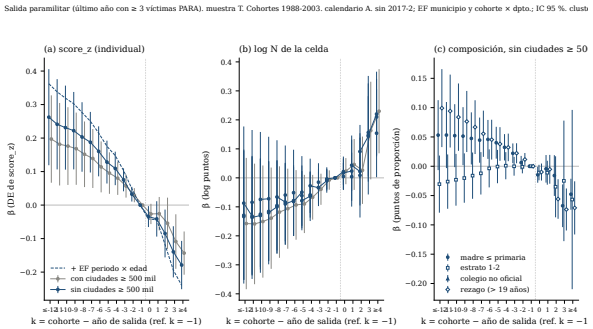
The standard design just failed its replication

Huaroto, Gonzales and Pacheco (2026, *JHR*): León (2012) with the full-count census.

- Same equation, same controls; 7 million observations instead of 139,000.
- Violence *before conception* has significant effects: the parallel pre-trends assumption fails.
- Only the in-utero effect survives Rambachan and Roth (2023, *REStud*) bounds.
- Cause: where violence arrives is correlated with historical determinants of human capital. The cross-sectional variation is not exogenous.

With our equation we would be proposing a design that has just been shown not to identify the age gradient it claims.

Our own data say the same



Saber 11, 3.2 million test takers, cohorts relative to the municipal end of violence. No break at the end: a monotone cohort trend, with composition and the number of test takers moving in parallel. The pre-conception placebo is significant on both the violence and the displacement clocks.

Migration makes exposure endogenous, and no public data fix it

- In affected municipalities the median share of population displaced in 1997–2010 is 23%.
- Exposure depends on whether the household stays and when it leaves; leaving is a different treatment.
- Municipality of birth fixes the location but not the problem: the intention-to-treat effect is a mixture of violence, displacement and the selection of who leaves, with weights we cannot recover.
- Separating channels needs individual migration histories linked to outcomes. They exist only in restricted linked records (ICFES, UARIV, PILA) with no access protocol.

The one escape route, and why it is not enough

Use the *end* of violence, set by a national negotiation calendar, as the source of variation.

- It is a different, plausibly more exogenous source than the arrival of violence. It is the only new ingredient.
- But 70% of exits fall within eighteen months (July 2004 to December 2005): little variation in the age at the end.
- Cohorts born after the end are not yet adults; with current data they exist in 28 municipalities.
- And the estimating equation would still be the 2012 one: two-way fixed effects with a continuous, staggered dose. Everything the recent difference-in-differences literature warns against.

Assessment

- **Question:** answered several times; its best-known answer has just been overturned, but not by us.
- **Design:** 2012 vintage; fails the one validity test that matters, in Peru and in our own exploratory data.
- **Mechanism:** not identifiable without linked individual records we do not have.
- **As formulated:** an incremental age-gradient paper with a 2012 design. Not viable.

Viable only if reformulated around a question that is actually open.

The open question, and what we keep

Is the long-run damage of violence prenatal or postnatal?

- After the 2026 replication, only the in-utero effect is credible. Whether exposure after birth matters at all is unknown.
- The dated end of paramilitary violence separates children exposed only in utero from those also exposed in infancy or preschool, within the same municipality of birth, with a source of variation set by a national negotiation calendar.

What we keep.

- A municipality $\times$ month $\times$ perpetrator panel of victims, 1985–2015 (CNMH), with entry, peak and exit dated for 451 municipalities.
- A displacement panel by municipality and year (RUV).
- A working pipeline for 9 million Saber 11 records with date and place.